David M. Plehn

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Profile

Second year at UC Irvine double-majoring in Computer Science and Quantitative Economics with hands-on experience in AI evaluation, data science, and peer-reviewed scientific research. Proven ability to build full-stack applications, analyze data with machine-learning techniques, and contribute to real-world AI systems. Seeking software engineering, data science, or AI internships.

Education

University of California, Irvine — Irvine, CA
B.S. in Computer Science (Intelligent Systems), B.A. in Quantitative Economics, Minor in Statistics — Sep. 2025 – June 2029
Relevant Coursework: Data Structures, Statistics, Linear Algebra, Differential Equations, Artificial Intelligence, Large Language Models

Technical Skills

Languages: Python, Java, C++, SQL, PostgreSQL, TypeScript/JavaScript, HTML/CSS, Dart, C#, LabVIEW, R, MIPS Assembly
Frameworks: PyTorch, TensorFlow/Keras, Scikit-Learn, Flutter, Vue.js, React Native
Developer Tools: Git, Google Colab, Jupyter Notebook, Visual Studio, PyCharm, IntelliJ, Firebase, FlutterFlow, Microsoft Office, AutoCAD, SonarQube
Libraries: Pandas, NumPy, Matplotlib, sqlite3, python-gitlab, Zustand, ExpoGo

Experience

Systems Engineer Intern — June 2026 – Present
Lockheed Martin — *Colorado Springs, CO*
- Automated mission-critical systems using Python scripting to enhance efficiency and reliability.
- Troubleshot deployment issues, ensuring seamless integration and modernization of network configurations.
- Applied DevOps principles, leveraging a software engineering background to solve complex challenges.
- Collaborated with cross-functional teams to enhance system security and reduce latency.

Artificial Intelligence Trainer — Nov. 2025 – Present
Handshake AI — *Irvine, CA*
- Designed AutoCAD Mechanical training projects to improve AI model accuracy and professional CAD workflow performance
- Built detailed scoring rubrics to evaluate AI recreation of CAD models, increasing output precision and consistency
- Developed and tested domain specific prompts to benchmark large language model performance in scientific and technical subjects
- Evaluated model responses for factual accuracy, reasoning quality, and clarity, driving measurable improvements in reliability
- Conducted independent research to strengthen prompt design, scoring criteria, and error analysis processes
- Delivered structured technical feedback to improve AI understanding of complex engineering and scientific concepts
- Trained and evaluated multimodal AI systems across text, audio, and visual inputs to increase response quality and cross modality accuracy

Research Assistant — Oct. 2024 – July 2025
University of Denver — *Denver, CO*
- Developed LabVIEW programs used for efficient and precise control of sensitive measurement equipment
- Collaborated on AutoCAD designs to build custom tools enhancing lab measurement capabilities
- Fabricated and validated thin-film devices using vacuum sputtering, lithography, and verification techniques for spintronics research
- Supported experimental workflows used in active physics research

Projects

GPT Model from Scratch | *PyTorch, Python, NumPy, Pandas* Jan. 2026 – Current

- Utilized Byte-Pair encoding to fine-tune NLP embeddings for initial vocabulary
- Implemented PyTorch's cuda architecture through WSL2 virtual environment to communicate with NVIDIA GPUs
- Designed attention model structure to construct core LLM logic
- Performed fine-tuning on outputs with a RLHF methodology

SceneCheck | *TypeScript, Supabase, JavaScript, HTML, PostgreSQL, Zustand, ExpoGo* March 2026 – June 2026

- Built a cross-platform web and mobile application enabling users to find, join, and create events
- Created recommendation algorithm to match events to users based on input interests
- Utilized Supabase for backend, authentication, data storage, and for PostgreSQL implementations
- Designed frontend data visualizations across platforms to highlight locations, attendees, descriptions, etc.
- Deployed Android, iOS, and web versions using ExpoGo

Co-Author of Peer-Reviewed Academic Research Paper in the *Journal of Applied Physics* Jan. 2025 – Sept. 2025

https://doi.org/10.1063/5.0300521

- Title: "Field-like torque measured via a Wheatsone bridge"
- For use in enabling reliable SOT-MRAM and superior process control and materials screening, designed an optimized Wheatstone bridge geometry that improved signal sensitivity for measuring field-like spin-orbit torque
- Enabled in-situ calibration of measurement devices, reducing experimental complexity while improving accuracy

Personal Achievements

Boy Scouts – Eagle Scout | Lockheed Martin CodeQuest Competition – 2nd Place| 1st Degree Black Belt in Karate